Machine Learning Lab Week 2

M.Tech. Mathematics and Computing

Topic: Learning by Probabilistic Modelling

(Basic Foundations, Classification/Regression, Naive Bayes, Linear & Logistic Regression, Generalized Linear Models)

General Instructions

- **All questions must be implemented in Python.**
- **You may use NumPy, Pandas, Matplotlib/Seaborn, and Scikit-learn unless stated otherwise.**
- **Clearly mention assumptions, observations, and interpretations.**
- **Wherever synthetic data is required, use a fixed random seed for reproducibility.**

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1. Generate a synthetic binary classification dataset using a Gaussian Mixture Model (two components, one per class). Train a Gaussian Naive Bayes classifier on the data.
    a. Compare the learned class-conditional distributions with the true generating distributions.
    b. Comment on where and why they differ.
2. Using a synthetic binary classification dataset (`make_classification`), train a Naive Bayes classifier with different class priors**.**
    a. Analyze how posterior probabilities and predictions change.
    b. Visualize decision boundaries for at least two different priors.
3. Create a dataset with highly correlated features**.**
    a. Train a Naive Bayes classifier and a Logistic Regression model.
    b. Compare their performance and explain the results using probabilistic assumptions.
4. Implement linear regression from scratch using Maximum Likelihood Estimation (MLE).
    a. Compare learned parameters with `sklearn.linear_model.LinearRegression`.
    b. Discuss numerical differences, if any.
5. Generate a regression dataset with heteroscedastic noise.Train a linear regression model.
    a. Compare Mean Squared Error (MSE) and log-likelihood based loss.
    b. Interpret results probabilistically.
6. Using a synthetic regression dataset:
    a) Implement linear regression using MLE.
    b) Implement MAP estimation assuming a Gaussian prior on parameters.
    c) Compare parameter estimates and predictive uncertainty.

7. Use a binary classification dataset (`make_classification`).

a. Train logistic regression models using different link functions (logit, probit).
b. Compare convergence behavior and classification performance.

8. Generate a non-linearly separable dataset (e.g., concentric circles or moons).
   a) Train a Logistic Regression model.
   b) Visualize decision boundaries.
   c) Explain probabilistically why performance is limited.

9. For a 2D linear regression problem:
   a) Compute and plot the likelihood surface over two parameters.
   b) Identify regions of high and low likelihood.

10. Train two probabilistic classifiers (e.g., Naive Bayes and Logistic Regression) on the same dataset.
    a) Compute the KL-divergence between their predicted probability distributions.
    b) Interpret which model is more informative.

11. Using a real dataset (Breast Cancer Wisconsin or similar):
    a. Train Naive Bayes and Logistic Regression models.
    b. Evaluate probability calibration using reliability diagrams and Brier score.

12. Using Bayesian linear regression:
    a) Derive and implement the posterior predictive distribution.
    b) Visualize prediction uncertainty bands.

13. Create a highly class-imbalanced dataset.
    a) Train a probabilistic classifier.
    b) Analyze how probability estimates become misleading.
    c) Apply Bayesian reweighting or class priors to correct this.

14. Generate a random dataset having two classes or load an existing dataset. Train: One classifier optimized for log-likelihood, One optimized for accuracy Compare probability confidence and decision behavior.

15. Using a classification dataset:
    a. Select features using entropy and mutual information.
    b. Compare results with correlation-based feature selection.

16. Implement GLMs using different exponential family distributions:
    a) Gaussian
    b) Poisson
    c) Binomial
    d) Gamma

Evaluate which distribution fits which dataset best and justify probabilistically.

17. Train a probabilistic density model on normal data.

    a) Detect anomalies using likelihood thresholding.
    b) Compare with a reconstruction-error-based approach.

18. Use a synthetic binary classification dataset with overlapping classes.
    a. Train a Logistic Regression model.
    b. Compare hard labels (threshold = 0.5) with probability outputs.
    c. Analyze prediction instability near the decision boundary.
19. Use a synthetic regression or classification dataset generated using `sklearn.datasets.make_regression` or `make_classification` (choose one).
    a. Fit two competing probabilistic models (e.g., linear vs polynomial regression, or Logistic Regression vs Naive Bayes).
    b. Compute the log-likelihood of each model on the same dataset.
    c. Perform a Likelihood Ratio Test (LRT) to statistically compare the models.
    d. Interpret the test statistic and decision at a chosen significance level.